

developing IoT projects, and has ultimately built an appropriate model.

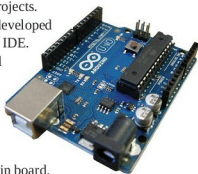
"Particularly in the MCU domain, there are millions of Arduino developers demanding advanced features to make their development with devices easier. Keeping open source and open platforms in mind, we started the work to add an extension on Visual Code Studio," said Zhidi Shang, R&D and product development, Microsoft.

While there are a number of tools for the cloud, interactive devices and gateways, the availability of tools for microcontrollers and sensors is very limited. Arduino developers have been demanding some tools to simplify the required development efforts while working on Arduino projects.

The simple extension of Visual Studio Code developed by Microsoft looks similar to the official Arduino IDE. The extension is fully compatible with the official Arduino IDE, and the missing advanced features are IntelliSense, Auto Code completion and on-device debugging.

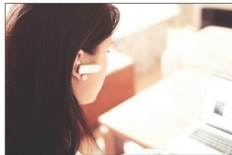
You can easily upload and verify your Arduino sketches in the Visual Studio Code. Further, the extension is fully loaded with a built-in board, example list, serial monitor, snippets for sketches and a library manager.

You can access the Visual Studio Code Extension for Arduino from its GitHub repository. A Gitter account is also available to let you join discussions with other developers and IoT enthusiasts to improve the development of your latest project.



Astecs launches IP PBX solution for SMEs

Astecs has expanded its line-up of enterprise telecom technologies by launching a new IP PBX solution specifically for SMEs. Called AstLite, the new offering is ideal for offices with up to 80 extensions.



"AstLite enables small businesses to take advantage of its high-impact features and deliver an enriched end-user experience," said Devasia Kurian, CEO of Astecs.

AstLite interoperates with traditional standards-based telephony and Voice-over-IP (VoIP) systems, company sources claim. The technology has open standards

with seamless scalability and easy-to-use management features. The Bengaluru-based company is offering 24x7 support alongside built-in recording, centralised administration and enhanced security to ensure real-time monitoring.

Based on the open source enterprise telecom solution Asterisk, AstLite has features such as smart telephony with a Wi-Fi client, audio conference bridging, interactive voice responses, voice loggers, a distributed office set-up, 'voice mail to email' and 'fax to email', among others. The solution also reduces telephony costs by prioritising and routing communications.

Oracle releases three open source container solutions

Continuing its efforts to expand its container offerings, Oracle has brought out three new open source utilities. The new developments are to help users improve the security of application containers.

The tools published by Oracle, which include the Smith Secure Container Builder, the Crashcart container debugging tool and the Railcar container runtime,

Open source MS-DOS alternative turns 23

FreeDOS, which was originally conceived as an open source alternative to MS-DOS, has completed 23 years. The community-powered project was launched back in 1994.

FreeDOS was first a small project that replaced MS-DOS, the default solution in most PCs in the 1980s and 1990s. But due to limited features for MS-DOS users, open source enthusiast and FreeDOS founder Jim Hall decided to extend its capabilities.

Microsoft planned to eliminate MS-DOS in the 1994 version of Windows; so Hall created his own version of DOS and launched it on June 22, 1994. The project debuted as PD-DOS (Public Domain DOS) and soon grew into FreeDOS. "We could create our own version of DOS, something that worked with programs meant for MS-DOS, but our DOS would be free for everyone to use," Hall wrote in a blog post, recalling the launch of FreeDOS.

Even after 23 years, FreeDOS is still popular and is used by the open source community around the world — to run business software, develop embedded systems or to play classic DOS games. Developers are also actively contributing to FreeDOS.

You can install the FreeDOS package on to your system and start coding using C, Pascal, Assembly, BASIC or plenty of other programming languages. The standard FreeDOS editor is easy to understand and intuitive.

The FreeDOS 1.2 distro released last December following a four-year release schedule, had 15 different code editors, the Dillo graphical Web browser and the Lynx text-mode Web browser. It also included interesting DOS games like Nethack, Beyond the Titanic, Paku Paku, Wing, a few flight simulators and other popular card games.